

Mohamed Martini

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Skills: Python, AI/ML (LangChain, LangGraph, TensorFlow, PyTorch), Cloud (GCP - Cloud Run, CloudSQL, Cloud Trace), Data (PostgreSQL, BigQuery, Apache Beam), DevOps (Docker, CI/CD), Web (FastAPI), Flutter.

PROFESSIONAL EXPERIENCE

Founder & Principal Engineer | Lever AI Full-time · July 2025 - Present

- Founded a consulting and SaaS development LLC specialized in AI-powered, data-driven software solutions.
- **Showcase Project: Guesstimate: Not Trivia!** ([Google Play](#), [App Store](#))
 - Designed and shipped a full-stack social estimation game to Android and iOS, featuring **real-time multiplayer** via Firestore, global leaderboards, and freemium monetization (RevenueCat, Unity Ads).
 - Built an **LLM-powered ETL pipeline** using GPT-4o and Gemini models for question generation, pgvector embeddings for **semantic deduplication**, **Google AI engine** for answering, and **semaphores** for mitigating API rate limits. Generated 1000+ high-quality, diverse, and fact-checked questions at <1 cent per question.
 - Achieved **high consistency** for a multi-player game mode using **atomic firestore transactions**.
 - Architected **low-latency**, fair question assignment using **optimized SQL** queries and indexing.
 - Implemented **end-to-end distributed tracing** with OpenTelemetry, correlating Flutter frontend events to FastAPI back-end spans via Cloud Trace and Cloud Logging.
 - Established **GitFlow CI/CD pipelines** with automated unit and integration testing, and dev/prod building and deployment.
 - Leveraged **AI coding agents** to accelerate cross-platform Flutter development while maintaining full architectural oversight of the Python backend.

Machine Learning Engineer III | Pison Technology Full-time · Oct 2022 - July 2025 · 2 yrs 10 mos

- **Promoted from MLE to MLE III** in April 2025.
- Ran 1,200+ TensorFlow experiments for sleep stage classification, improving the baseline F1-score by 15%. Developed **two custom loss functions** to address class imbalance. Designed and Implemented a **custom attention layer** for feature mining, improving performance while reducing the size of the feature set by an order of magnitude.
- Developed real-time biosignal gesture detection models. Wrote Cythonized extension modules, and vectorized bottleneck functions, significantly improving the speed of online prediction and offline modeling.
- Led a “Hackaweeek” project with two colleagues to implement a state-of-the-art [domain-invariant classification network with adversarial training](#) in PyTorch, following the approach outlined in its original manuscript.
- Led the adoption of BigQuery as a data warehouse solution to streamline the handling of analytics data.
- Wrote a scalable Batch ETL pipeline using Apache Beam which brought down the compute time of 56 GB of data from 2 days to 15 minutes at a cost of 25 cents.
- Standardized the Python codebase, CI/CD workflows, and dependency management, enabling seamless collaboration between colleagues across operating systems.
- Maintained cross-team communication to ensure consistency and optimal integration of ML solutions.

Research Assistant | UNIVERSITY OF MASSACHUSETTS LOWELL Jan 2021 – June 2022

- Implemented **Reinforcement Learning** agents for cooperative multi-agent Search and Rescue simulations.
- Implemented **object detection** training and inference pipelines. Used Generative AI for data augmentation.

EDUCATION

- **M.S. Computer Engineering** · GPA 4.0 · UMass Lowell, 2022
- **B.S. Electrical Engineering** · GPA 3.8 · UMass Lowell, 2021

REFERENCES

References available upon request.

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